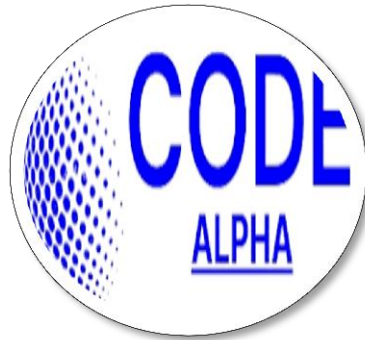


CODE ALPHA
Empowering Future Innovators



INTERNSHIP REPORT

On
BIOINFORMATICS

Task 2: MULTIPLE SEQUENCE ALIGNMENT

Submitted by:

Faiqa Riaz

Internship Duration: 1 Month

Submitted to: Code Alph

PROTEIN STRUCTURE PREDICTION

AlphaFold-Predicted 3D Structure of Human Insulin

1. Introduction

Predicting the three-dimensional structure of a protein from its amino acid sequence is one of the central problems in structural biology, since a protein's shape largely determines its function. AlphaFold, developed by Google DeepMind, is a deep-learning system that predicts protein structures with an accuracy that in many cases rivals experimental methods such as X-ray crystallography, and its predictions for essentially the entire UniProt database are freely available through the AlphaFold Protein Structure Database.

This report examines the AlphaFold-predicted structure of human insulin (UniProt: P01308), continuing the analysis of this protein from the sequence-level work carried out in Tasks 1 and 2. The aim is to visualize the predicted 3D structure, interpret AlphaFold's per-residue confidence scores, and relate the structural features observed to insulin's known biology.

2. Methodology

- The UniProt accession P01308 (human insulin) was searched in the AlphaFold Protein Structure Database (alphafold.ebi.ac.uk).
- The search initially returned a multimeric complex (insulin bound to IGF-binding protein 4); the results were filtered to "Monomers" to isolate the standalone AlphaFold model for insulin alone (entry AF-P01308-F1).
- The interactive 3D structure viewer was used to rotate the model and inspect it from multiple angles.
- The per-residue confidence (pLDDT) colouring and the Predicted Aligned Error (PAE) plot provided on the same page were reviewed to assess which regions of the model are reliable.

Table 1. Summary information for the AlphaFold model used in this analysis.

Field	Value
Protein	Insulin

Gene	INS
UniProt accession	P01308
Source organism	Homo sapiens
Sequence length	110 residues (precursor)
AlphaFold model	AF-P01308-F1, Version 6
Global quality	Average pLDDT 52.91 (Low)

3. Results

Figure 1 shows the full AlphaFold results page for P01308, including the sequence, the 3D structure viewer, the pLDDT confidence legend, and the Predicted Aligned Error (PAE) matrix. Figure 2 shows the same structure rotated to a side-on view that more clearly separates its component parts.

Insulin

[Download files](#)

Monomer • AF-P01308-F1 • Version v6 • Google DeepMind dataset • 

Summary and Model Confidence

[Domains](#)
[Annotations](#)
[Similar Proteins](#)

Protein	Insulin	Found in	3 entries in AFDB go to search
Gene	INS	Experimental structures	386 structures in PDB for P01308 go to PDB-KB
Source organism	<i>Homo sapiens (Human) (Human)</i> go to search	Average pLDDT	52.91 (Low)
UniProt	P01308 go to UniProt	pLDDT distribution	0% Very high 12.7% High 36.4% Low 50.9% Very low
Biological function	Insulin decreases blood glucose concentration. It increases cell permeability to monosaccharides, amino acids and fatty acids. It accelerates glycolysis, the ... Show more go to UniProt		

[Copy sequence](#)

A | 1: Insulin

1 11 21 31 41 51 61 71 81 91 101
 MALWMRLFLPALLALMWGPDAFAAFVNHQGHSHIVEALYIVCGERGFFTFKTRREARDLQVGVQLGGGPGAGSLQPLALEGSSIQKRGIVQQCTSIQSILYQLENYCN

Activate Windows

Go to Settings to activate Windows



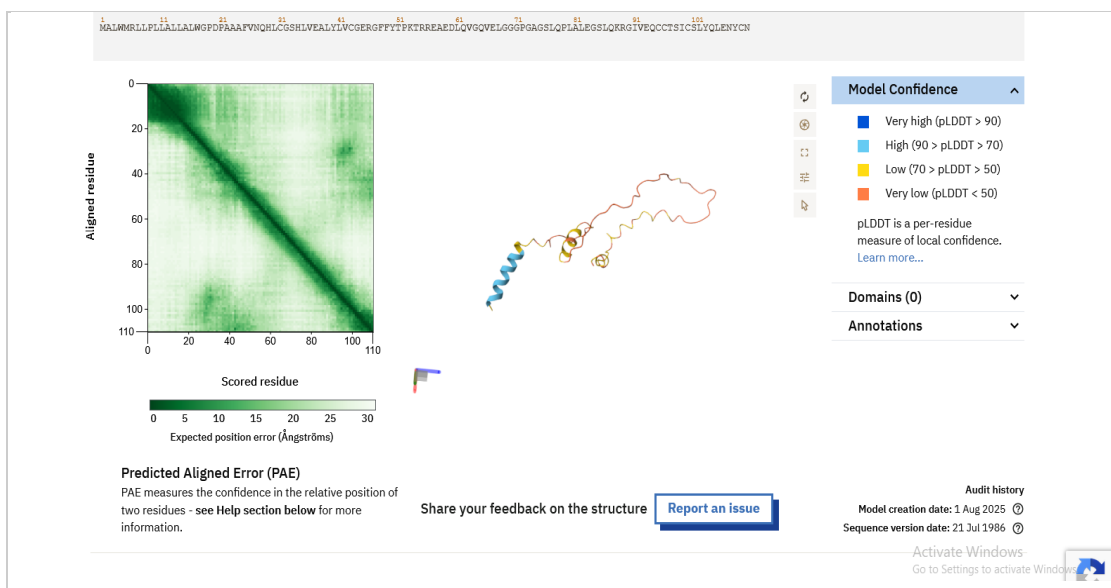


Figure 1. AlphaFold structure page for human insulin (AF-P01308-F1), showing the query sequence, the 3D model coloured by per-residue confidence (pLDDT), the Model Confidence legend, and the



Predicted Aligned Error (PAE) plot.

Figure 2. Rotated view of the same model, showing the blue alpha-helix, the short yellow helical segments, and the long, loosely coiled orange/red region.

The model shows two short alpha-helices — one coloured blue (high confidence, pLDDT 70-90) and a pair of shorter helices coloured yellow (low confidence, pLDDT 50-70) — connected by a long, largely unstructured coil rendered in orange and red (very low confidence, pLDDT < 50). The blue helix corresponds to part of the B-chain, while the yellow helical segments correspond to the A-chain; the long red/orange coil in between is the C-peptide, which is enzymatically removed when the precursor is processed into mature, active insulin.

4. Interpretation

- Confidence pattern reflects biology, not model error: The low overall average pLDDT (52.91) is expected rather than a sign of a poor prediction. AlphaFold's confidence score reflects how consistently a region folds into a fixed structure across its training data. The C-peptide is known to be intrinsically disordered in isolation — it only exists transiently in the precursor and has no fixed 3D shape of its own, so AlphaFold correctly reports it with very low confidence.
- The functional core is high-confidence: The B-chain and A-chain segments, which form the compact, disulfide-linked structure of mature insulin, are predicted with higher confidence (blue and yellow) than the C-peptide. These are the regions that actually fold into a defined shape and bind the insulin receptor.
- PAE plot supports this reading: The Predicted Aligned Error matrix (Figure 1, left) shows a tight, dark-green diagonal band for residues 1-30 and 90-110 approximately, indicating AlphaFold is confident about the relative positions of atoms within the B-chain and A-chain regions, while the pale, low-confidence area covering the middle of the plot corresponds to the flexible C-peptide, whose position relative to the rest of the structure is essentially unpredictable — consistent with it being cleaved away in the biologically active hormone.
- Biological interpretation: This structure represents the full pre-processed insulin precursor, not the mature two-chain hormone seen in most textbook diagrams. In the body, the C-peptide is enzymatically cut out, and the resulting A-chain and B-chain remain joined by disulfide bonds to form the active insulin molecule. AlphaFold's prediction is therefore best read as showing which parts of the precursor are structurally meaningful (B-chain, A-chain) and which are just an unstructured linker (C-peptide) that plays no role in the final, functional protein.

5. Conclusion

The AlphaFold-predicted structure of human insulin (P01308) shows two short, well-defined helical segments corresponding to the B-chain and A-chain, connected by a long, disordered C-peptide region. Rather than indicating a poor-quality model, the low overall confidence score correctly reflects the intrinsically disordered nature of the C-peptide in the unprocessed precursor. This exercise demonstrates how

AlphaFold's per-residue confidence metrics (pLDDT and PAE) can be used not just to judge model reliability, but to distinguish structurally and functionally important regions of a protein from flexible linker regions — reinforcing, at the structural level, the same B-chain/A-chain versus C-peptide distinction that emerged from the sequence conservation patterns observed in Task 2.

References

Jumper, J. et al. (2021). Highly accurate protein structure prediction with AlphaFold. Nature, 596, 583-589.

AlphaFold Protein Structure Database (Google DeepMind / EMBL-EBI). Available at: <https://alphafold.ebi.ac.uk/>

UniProt Consortium. UniProt Knowledgebase (UniProtKB). Available at: <https://www.uniprot.org/>